

The (s, t) -square closed: Hikita's $\star$ on all four edges

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Title: The (s, t) -square closed: Hikita's $\star$ on all four edges
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Source: proofs/2026-10-02-day217e-boundary-of-the-st-square.md; scripts in scripts/day217e/
Grade: Lemma R, Theorems A, B, Prop. C, Cor. D: *proved* (self-proved, not peer-reviewed), conditional on (N) (Day 216c; your review pending) and on Macdonald-book locators recalled from memory. Machine-checked in small cases (§8).
Novelty: (N) itself is folklore-implicit (Di Francesco–Kedem, arXiv:1704.00154). What is claimed here is only the explicit edge identifications in Hikita's $\star$ -monomial basis. Theorem B in particular may be known in disguise; see §9.

Summary. Theorems H (edge $s = 0$) and H' (edge $t = \infty$) described $\star$ through the family $\Psi(b_\mu)$. That family *collapses* onto e_n on the other two edges. The missing edges are carried by the other family, the $\star$ -monomials $e_\mu^\star = e_{\mu_1} \star \cdots \star e_{\mu_\ell}$. The swap is a symmetry: $P(x; q, t) = P(x; 1/q, 1/t)$ gives $\Psi_{1/s, 1/t} = \Psi_{s, t}^{-1}$. The point reflection of the square through $(1, 1)$ therefore sends H to a new Theorem A ($s = \infty$: Hall–Littlewood $P_{\mu'}(x; t)$) and H' to a new Theorem B ($t = 0$: $e_\mu^\star = \omega \tilde{H}_\mu(x; s)$, cocharge Kostka–Foulkes).

1 Setup

As in the (N) PDF: $s = q^{-1}$, and Hikita's operators are

$$E_k F = \sum_{|A|=k} \prod_{i \in A, j \notin A} \frac{x_i - tx_j}{x_i - x_j} X_A F(X_{A^c}, sX_A), \quad e_k \star F := E_k F.$$

- $e_\mu^\star := e_{\mu_1} \star \cdots \star e_{\mu_\ell} = E_{\mu_1} \cdots E_{\mu_\ell}(1)$ ($\star$ is commutative).
- $b_\mu := s^{n(\mu)} e_\mu$ (ordinary product).
- $\Psi = \Psi_{s, t} : (\text{Sym}, \star) \rightarrow (\text{Sym}, \cdot)$ is the algebra isomorphism with $\Psi(e_\mu^\star) = t^{-n(\mu')} e_\mu$.
- $P_\nu := P_\nu(x; s, 1/t)$ (Macdonald), $T_\nu := t^{n(\nu)} s^{n(\nu')}$, and $\mathcal{N}P_\nu = T_\nu P_\nu$.
- (N) (Day 216c): $\Psi = \mathcal{N}^{-1}$.

Standard facts (Macdonald, SFHP 2nd ed.; locators from memory).

- (M1) $P_\nu(x; q, t) = P_\nu(x; q^{-1}, t^{-1})$ [VI (4.14)(iv)].
(M2) $P_\nu(x; 0, t) = P_\nu(x; t)$ (HL); $P_\nu(x; q, 1) = m_\nu$; $P_\nu(x; 1, t) = e_{\nu'}$; $P_\nu(x; q, q) = s_\nu$ [VI (4.14)].
(M3) *Regularity.* By the tableau formula VI (7.13'), the m -coefficients of P_ν are sums of products of factors $b_\mu(s)/b_\lambda(s)$, each equal to

$$\frac{(1 - q^{a_\mu} t^{l_\mu + 1})(1 - q^{a_\lambda + 1} t^{l_\lambda})}{(1 - q^{a_\mu + 1} t^{l_\mu})(1 - q^{a_\lambda} t^{l_\lambda + 1})}.$$

At $q = 0$ each denominator factor becomes 1 or $1 - t^{l+1}$; at $t = 0$ it becomes 1 or $1 - q^{a+1}$.

So P_ν is regular at $q = 0$ (generic t) and at $t = 0$ (generic q).

- (M4) *Unitriangularity.* $P_\nu = m_\nu + \sum_{\kappa \triangleleft \nu} (\cdots) m_\kappa$ and $e_\mu = m_{\mu'} + \sum_{\kappa \triangleleft \mu'} (\cdots) m_\kappa$. Hence $e_\mu = \sum_{\nu \trianglelefteq \mu'} \alpha_{\mu\nu} P_\nu$ with $\alpha_{\mu\mu'} = 1$, and by (M3) each $\alpha_{\mu\nu}(q, t)$ is regular at $q = 0$ and at $t = 0$.
(M5) $P_\lambda(x; q, 0) = \omega Q'_\lambda(x; q)$, where $Q'_\mu(x; q) = \sum_\lambda K_{\lambda\mu}(q) s_\lambda$ (Day 216b §4, from VI (5.1)).
Finally, $n(\cdot)$ is strictly order-reversing on dominance: $\nu \triangleleft \kappa \Rightarrow n(\nu) > n(\kappa)$.

2 The reflection lemma

Lemma R (reflection). $\Psi_{1/s, 1/t} = \Psi_{s, t}^{-1}$; equivalently $\mathcal{N}_{1/s, 1/t} = \mathcal{N}_{s, t}^{-1}$.

Proof. $\mathcal{N}_{1/s, 1/t}$ is diagonal on $P_\nu(x; 1/s, t)$, which equals $P_\nu(x; s, 1/t)$ by (M1). Its eigenvalue is $T_\nu(1/s, 1/t) = t^{-n(\nu)} s^{-n(\nu')} = T_\nu(s, t)^{-1}$. Now apply (N) at both points. $\square$

R-duality. Using $\mathcal{N}(e_\mu) = t^{n(\mu')} e_\mu^*$:

$$\Psi_{s, t}(b_\mu) = s^{n(\mu)} \mathcal{N}_{1/s, 1/t}(e_\mu) = s^{n(\mu)} t^{-n(\mu')} e_\mu^*|_{(1/s, 1/t)}.$$

So the reflection $R : (s, t) \mapsto (1/s, 1/t)$ exchanges the edges $s = 0 \leftrightarrow s = \infty$ and $t = \infty \leftrightarrow t = 0$. It also exchanges the families $b \leftrightarrow e^*$.

Master function. Put $F_\mu(s, t) := \mathcal{N}_{s, t}(e_\mu) = \sum_{\nu \trianglelefteq \mu'} \alpha_{\mu\nu} T_\nu P_\nu$. Then

$$e_\mu^*(s, t) = t^{-n(\mu')} F_\mu(s, t), \quad \Psi_{s, t}(b_\mu) = s^{n(\mu)} F_\mu(1/s, 1/t).$$

3 Termwise valuation

On the support $\nu \trianglelefteq \mu'$ we have

$$T_\nu/T_{\mu'} = t^{n(\nu) - n(\mu')} s^{-(n(\mu) - n(\nu'))}.$$

Both exponents are ≥ 0 , and each vanishes iff $\nu = \mu'$. Consequently:

- As $t \rightarrow 0$ or $s \rightarrow \infty$, the term $\nu = \mu'$ dominates *uniquely*. These are the good edges for F_μ .
- As $t \rightarrow \infty$ or $s \rightarrow 0$, the term $\nu = 1^n$ dominates uniquely. These are the collapse edges, provided $\alpha_{\mu, 1^n} \neq 0$ there (Prop. C).

For $\Psi(b_\mu)$ the roles flip: $s = 0$ and $t = \infty$ are good (Theorems H, H'), while $s = \infty$ and $t = 0$ collapse.

4 Theorem A: the $s = \infty$ edge

Theorem A ($s = \infty$ edge). *For every μ , as $s \rightarrow \infty$ with t generic and fixed,*

$$s^{-n(\mu)} e_{\mu_1} \star \cdots \star e_{\mu_\ell} \longrightarrow P_{\mu'}(x; t) \quad (\text{Hall-Littlewood } P, \text{ parameter } t).$$

Proof. 1. Let $q = 1/s$. By (M1), $P_\nu = P_\nu(x; q, t)$ and $\alpha_{\mu\nu} = \alpha_{\mu\nu}(q, t)$, which is regular at $q = 0$ by (M4).

2. $e_\mu^* = \sum_{\nu \leq \mu'} \alpha_{\mu\nu}(q, t) t^{n(\nu) - n(\mu')} s^{n(\nu')} P_\nu(x; q, t)$.

3. Multiply by $s^{-n(\mu)}$. The ν -term carries $s^{n(\nu') - n(\mu)}$, which tends to 0 for $\nu \neq \mu'$ (the exponent is ≤ -1 by §3) and equals 1 at $\nu = \mu'$. All other factors stay bounded as $q \rightarrow 0$, by (M3) and (M4).

4. The limit is $\alpha_{\mu\mu'}(0, t) P_{\mu'}(x; 0, t) = P_{\mu'}(x; t)$, by (M4) and (M2). $\square$

Remark. Via R-duality, Theorem A at (s, t) is literally Theorem H at $(1/s, 1/t)$.

5 Theorem B: the $t = 0$ edge

Theorem B ($t = 0$ edge). *For every μ , e_μ^* is regular at $t = 0$, and*

$$e_{\mu_1} \star \cdots \star e_{\mu_\ell} \Big|_{t=0} = s^{n(\mu)} P_{\mu'}(x; 1/s, 0) = s^{n(\mu)} \omega Q'_\mu(x; 1/s) = \sum_\lambda \tilde{K}_{\lambda\mu}(s) s_{\lambda'} = \omega \tilde{H}_\mu(x; s).$$

Here $\tilde{K}_{\lambda\mu}(q) = q^{n(\mu)} K_{\lambda\mu}(1/q)$ is the cocharge Kostka-Foulkes polynomial and $\tilde{H}_\mu(x; q) = \sum_\lambda \tilde{K}_{\lambda\mu}(q) s_\lambda$ is the modified Hall-Littlewood function.

Proof. 1. As in Theorem A with $q = 1/s$: $e_\mu^* = \sum_{\nu \leq \mu'} \alpha_{\mu\nu}(q, t) t^{n(\nu) - n(\mu')} s^{n(\nu')} P_\nu(x; q, t)$.

2. For generic s , both α and P are regular at $t = 0$ (M3, M4). The t -exponents $n(\nu) - n(\mu')$ are ≥ 0 , with equality only at $\nu = \mu'$ (§3).

3. At $t = 0$ this gives $e_\mu^* = \alpha_{\mu\mu'}(q, 0) s^{n(\mu)} P_{\mu'}(x; q, 0) = s^{n(\mu)} P_{\mu'}(x; 1/s, 0)$.

4. By (M5), $P_{\mu'}(x; q, 0) = \omega Q'_\mu(x; q) = \sum_\lambda K_{\lambda\mu}(q) s_{\lambda'}$. Substitute $q = 1/s$ and multiply by $s^{n(\mu)}$. $\square$

Remark. Via R-duality, Theorem B at (s, t) is Theorem H' at $(1/s, 1/t)$.

Examples. $e_1 \star e_1 \Big|_{t=0} = s h_2 + e_2 = s_{11} + s s_2$ (checked by hand from $\mathcal{N}$). At $s = 1$: $\sum_\lambda K_{\lambda\mu} s_{\lambda'} = e_\mu$, consistent with $\star = \cdot$ at $s = 1$.

6 Proposition C: the collapse edges

Proposition C (collapse coefficient). *For all (q, T) ,*

$$[P_{1^n}(x; q, T)] e_\mu = \left[\begin{matrix} n \\ \mu_1, \dots, \mu_\ell \end{matrix} \right]_T \frac{\prod_i (q; T)_{\mu_i}}{(q; T)_n} =: c_\mu(q, T).$$

Proof. 1. By orthogonality (VI (4.7)), $[P_{1^n}]f = \langle f, P_{1^n} \rangle_{q, T} / \langle P_{1^n}, P_{1^n} \rangle_{q, T}$, and $P_{1^n} = e_n$ by (M2).

2. $\langle \cdot, \cdot \rangle_{q, T}$ is diagonal on power sums, with weights multiplicative in the parts, and the p_r are primitive. So $\langle fg, h \rangle = \langle f \otimes g, \Delta h \rangle$.

3. Since $\Delta e_n = \sum_{a+b=n} e_a \otimes e_b$, degree matching gives $\langle e_\mu, e_n \rangle = \prod_i \langle e_{\mu_i}, e_{\mu_i} \rangle$.

4. $\langle e_r, e_r \rangle = 1/b_{1r}(q, T)$ with $b_{1r} = (T; T)_r / (q; T)_r$ (VI (4.11), (6.19)).

5. Hence $c_\mu = \prod_i [(q; T)_{\mu_i} / (T; T)_{\mu_i}] \cdot (T; T)_n / (q; T)_n$. $\square$

Specialisations: $c_\mu(0, T) = \begin{bmatrix} n \\ \mu \end{bmatrix}_T$; $c_\mu(q, 0) = (1 - q)^{\ell(\mu)-1}$; $c_\mu(1, T) = 0$ for $\ell \geq 2$.

Corollary C' (the four collapse limits). • $(s = \infty) s^{-n(\mu)} \Psi(b_\mu) \rightarrow t^{-\binom{n}{2}} \begin{bmatrix} n \\ \mu \end{bmatrix}_t e_n$.

- $(t = 0) t^{\binom{n}{2}} \Psi(b_\mu) \rightarrow s^{n(\mu)-\ell(\mu)+1} (s-1)^{\ell(\mu)-1} e_n$.
- $(s = 0) e_\mu^*|_{s=0} = \begin{bmatrix} n \\ \mu \end{bmatrix}_t e_n$. In particular $e_a \star e_r|_{s=0} = \begin{bmatrix} a+r \\ a \end{bmatrix}_t e_{a+r}$; this is the $q \rightarrow \infty$ Gaussian limit, matching Hikita Thm C(ii).
- $(t = \infty) t^{n(\mu')-\binom{n}{2}} e_\mu^* \rightarrow (1-s)^{\ell(\mu)-1} e_n$.

Proof. Each limit follows from §3 (unique dominant term $\nu = 1^n$) together with Prop. C, using (M3)/(M4) regularity in the relevant parameter. For example, as $s \rightarrow \infty$, with $q = 1/s$,

$$\Psi(b_\mu) = \sum_\nu \alpha_{\mu\nu}(q, t) s^{n(\mu)-n(\nu')} t^{-n(\nu)} P_\nu(x; q, t).$$

The top s -power is carried by $\nu = 1^n$, with coefficient $\alpha_{\mu, 1^n}(0, t) = \begin{bmatrix} n \\ \mu \end{bmatrix}_t$. $\square$

So the b -family is rank one on $s = \infty$ and $t = 0$, and the $\star$ -monomial family is rank one on $s = 0$ and $t = \infty$. Neither family sees the whole square; together they do.

7 Corollary D: operator form (b) of H'

Corollary D (operator form (b) of H'). Let $M_{\nu\mu}^{(k)}(s, t) := [b_\nu] E_k b_\mu$. Then $\deg_t M_{\nu\mu}^{(k)} \leq n(\nu') - n(\mu') - \binom{k}{2}$. The coefficient of $t^{n(\nu')-n(\mu')-\binom{k}{2}}$ is $\psi_{\nu/\mu}(s)$, the HL Q -Pieri coefficient (Macdonald III (5.7)–(5.8)), when ν/μ is a horizontal k -strip, and is 0 otherwise.

Proof. 1. Let $\beta_\mu(t) := t^{n(\mu')} \Psi(b_\mu)$. Applying Ψ to $E_k b_\mu = \sum_\nu M_{\nu\mu} b_\nu$, and using $\Psi E_k = t^{-\binom{k}{2}} e_k \Psi$, gives

$$e_k \beta_\mu = \sum_\nu M_{\nu\mu} t^{\binom{k}{2}+n(\mu')-n(\nu')} \beta_\nu.$$

2. By H', $\beta_\nu(t) \rightarrow \omega Q'_\nu(x; s)$ as $t \rightarrow \infty$, and these limits form a basis. So the coordinate functionals converge too (the inverse of a rational matrix converging to an invertible one converges).
3. Hence $M_{\nu\mu} t^{\binom{k}{2}+n(\mu')-n(\nu')} \rightarrow [Q'_\nu] h_k Q'_\mu$.
4. From $Q_\mu q_k = \sum_\nu \psi_{\nu/\mu}(s) Q_\nu$ (III (5.7)) and $q_k = h_k[(1-s)X]$, the plethysm $X \mapsto X/(1-s)$ gives $h_k Q'_\mu = \sum_\nu \psi_{\nu/\mu}(s) Q'_\nu$.
5. A rational $f(t)$ with $t^{-d} f(t)$ convergent as $t \rightarrow \infty$ has $\deg f \leq d$, and its t^d coefficient is the limit. $\square$

This upgrades the operator form (b) of H' from computed (136/136, $n+k \leq 5$) to proved. It also explains why the P -Pieri guess φ failed: the limit basis is Q' , not P .

8 Square Theorem (assembly) and verification

line	$\star$ becomes	basis statement	source
$s = 0$	HL product (param $1/t$)	$\Psi(b_\mu) = t^{-n(\mu')} P_{\mu'}(x; 1/t)$	Thm H
$t = \infty$	q -Whittaker (param s)	$t^{n(\mu')} \Psi(b_\mu) \rightarrow P_{\mu'}(x; s, 0) = \omega Q'_\mu(x; s)$	Thm H'
$s = \infty$	HL product (param t)	$s^{-n(\mu)} e_\mu^\star \rightarrow P_{\mu'}(x; t)$	Thm A
$t = 0$	modified HL / cocharge	$e_\mu^\star = \omega H_\mu(x; s) = s^{n(\mu)} P_{\mu'}(x; 1/s, 0)$	Thm B
$s = 1$	ordinary product	$e_\lambda \star e_\mu = e_{\lambda \cup \mu}$	DS / (N)
$t = 1/s$	content-twisted LR	$s_\lambda \star s_\mu = \sum c_{\lambda\mu}^\nu s^{c(\nu)-c(\lambda)-c(\mu)} s_\nu$	(N), Day 216
$t = 1$	$n(\cdot')$ -twisted monomial	$m_\lambda \star m_\mu = \sum_\nu [m_\nu] (m_\lambda m_\mu) s^{n(\nu')-n(\lambda')-n(\mu')} m_\nu$	(N)+(M2)

The reflection R swaps $H \leftrightarrow A$ and $H' \leftrightarrow B$, and it preserves the lines $s = 1$, $t = 1$ and $t = 1/s$. The corners are consistent. For example, at $(\infty, 0)$ with $\mu = (1, 1)$: $s^{-1}(e_1 \star e_1)|_{t=0} = h_2 + s^{-1}e_2 \rightarrow s_2 = P_2(x; 0)$.

Verification log. Scripts are in `scripts/day217e/`. The “Hikita” data are computed directly from the operators E_k ; the Macdonald P come from an independent Gram–Schmidt.

claim	script	range / result
Thm A	<code>estar_edges.py</code> , <code>estar_edges_num.py</code>	all $\mu \vdash n \leq 4$ symbolic (11/11); $n = 5$ exact at $t = 3/7$ (7/7)
Thm B	same	$n \leq 4$ symbolic (11/11); $n = 5$ exact at $s = 5/3$ (7/7)
negative control: A with $P_{\mu'}(x; 1/t)$	<code>estar_edges.py</code>	fails on (11), (21), (111), as it should
Cor. C', $e^\star$ collapse at $s = 0$, $t = \infty$	<code>estar_collapse.py</code>	$n \leq 5$: 36/36
Prop. C	<code>column_coeff.py</code>	$n \leq 4$ symbolic: 11/11
Cor. C', $\Psi(b)$ collapse at $s = \infty$, $t = 0$	<code>edges.py</code>	$n \leq 5$: 36/36 (exponent and coefficient)
Cor. D	<code>testb2.py</code> (Day 216)	$n + k \leq 5$: 136/136

9 Honesty: gaps and novelty

- Everything rests on (N), which is proved (Day 216c) modulo textbook Cherednik facts. **Your review of the (N) PDF is still pending.**
- The Macdonald locators (VI (4.14)(iv), (7.13'), (4.7), (4.11), (6.19); III (5.7), (2.10)) were recalled from memory and **not** re-verified this session. The (M3) regularity argument depends on the shape of the $b_\lambda(s)$ factors. As an independent sanity check, the computed limits exist and match for $n \leq 5$.
- **Novelty.** (N) is folklore-implicit: Di Francesco–Kedem (arXiv:1704.00154) have Hikita-type E_k as their $M_{k,1}$ in a ∇ -conjugation dictionary. Theorems A and B are R -reflections of H and H', so they inherit that status. The only claim here is the *explicit* edge identifications in Hikita's $\star$ -monomial basis, together with the collapse coefficient (Prop. C) and Cor. D.
- Theorem B looks like a cousin of Jing / Garsia-type creation operators for modified Hall–Littlewood functions. At $t = 0$, $E_k = \sum_A \prod_{i \in A, j \notin A} \frac{x_i}{x_i - x_j} X_A T_{s,A}$. A literature audit by operator formula is still to be done.

10 Question for you

Have you seen Theorem B in the ∇ form, namely that the normalised ∇e_μ at $t = 0$ (or $q = 0$) equals $\omega_{\tilde{H}_\mu}$, anywhere (Garsia–Haiman school, Haglund’s book)? Through (N) and the DFK dictionary, Theorem B reads “ T -normalised ∇e_μ at the degenerate parameter equals $\omega_{\tilde{H}_\mu}$ ”. If you have seen it, Theorems A and B are restatements, and any write-up should lead with the identification rather than the theorems.